Predicting and Explaining Employee Absenteeism at Work

Jennifer Huang, Shih-Hui Huang, Tse-Chun Hsu, Yu-Chen Lu

I. Problem Definition and Description of Background

Employee absenteeism is an important workforce issue because frequent or prolonged absences can disrupt operations, reduce productivity, and increase labor costs. This project analyzes the factors influencing absenteeism and builds predictive models to estimate absence time in hours, with a focus on demographic, work-related, transportation, lifestyle, and medical factors associated with longer absences. Prior studies have used the Absenteeism at Work dataset to predict absence duration with neuro-fuzzy models (Martiniano et al., 2012), data mining techniques (Skorikov et al., 2020), and tree-based machine learning methods (Wahid et al., 2019). Building on this work, we compare regression and classification approaches, consolidate absence reasons into broader interpretable groups, and apply clustering to identify employee-level absenteeism profiles.

II. Description of Dataset

We use the Absenteeism at Work dataset (Martiniano et al., 2012) from UCI, which contains 740 records and 21 attributes collected from a courier company in Brazil from July 2007 to July 2010. The target variable is absenteeism time in hours. The predictor variables capture a broad range of factors, including reasons for absence, temporal information, transportation, work history, age, workload, education, family characteristics, health-related measures, and lifestyle habits such as drinking and smoking. The dataset does not contain any missing values.

Strategic Refinement of Absence Reasons

The original dataset categorized reasons for absence into 21 International Classification of Diseases (ICD) codes and 7 special categories. To enhance interpretability and reduce sparsity, we consolidated the original absence reasons into 6 broader groups. Figure 1 shows the absenteeism patterns across our newly defined absence groups.

Medical Visits and *Other Medical* show a clear spike between 2 and 4 hours, indicating that a large proportion of absences are short-term and routine. In contrast, *Musculoskeletal* and *Injury*-related absences are less frequent but tend to last longer, often reaching full-day

[1] *Absenteeism at Work*, UCI Machine Learning Repository, https://archive.ics.uci.edu/dataset/445/absenteeism+at+work

durations (8, 16, or 24 hours). The *Other* and *Administrative* categories are more heterogeneous and do not exhibit a consistent pattern.

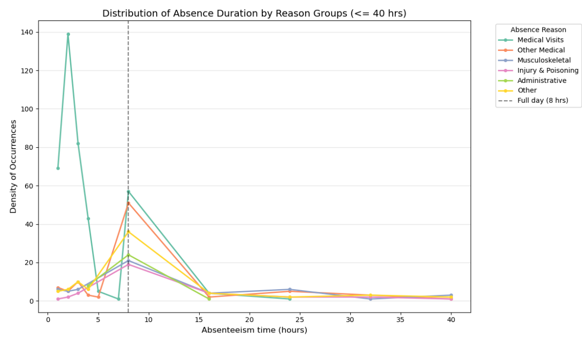

Figure 1. Distribution of Absence Duration by Reason Groups (<= 40 hrs)

Descriptive Statistics

The following descriptive statistics summarize the key demographic, work-related, lifestyle, and absenteeism-related variables in the dataset.

Table 1. Descriptive Statistics

Variable	All Samples	< 8 Hours	>= 8 Hours
AVERAGE ABSENCE HOURS	7.36 (13.63)	2.40 (1.04)	15.14 (19.41)
Transportation expense	219.92 (65.80)	206.06 (56.70)	241.65 (72.94)
Distance from Residence to Work	29.85 (14.86)	30.39 (14.99)	29.01 (14.63)
Service time	12.55 (4.44)	12.78 (4.42)	12.20 (4.46)
Work load Average/day	271.17 (39.10)	268.04 (36.40)	276.06 (42.59)
Son	1.00 (1.10)	0.84 (1.05)	1.24 (1.13)
Pet	0.73 (1.28)	0.65 (1.15)	0.86 (1.45)
Body mass index	26.57 (4.19)	26.53 (4.21)	26.64 (4.18)
Age: <30	24.70%	28.20%	19.20%
Age: 31-40	57.50%	56.20%	59.40%
Age: 41-50	16.70%	14.80%	19.60%
Age: >50	1.10%	0.70%	1.80%
Social drinker (%)	56.20%	52.00%	62.70%
Social smoker (%)	6.60%	4.90%	9.20%
Total Records (n)	696	425	271

Analysis of absence reasons and descriptive statistics reveals clear differences between short- and long-term absences, while demographic and work-related factors differ only slightly.

Short-term absences are driven by routine medical visits, whereas long-term absences are associated with structural factors such as domestic responsibilities and commuting burden, indicating distinct underlying mechanisms. Consequently, predicting exact absence hours is challenging due to high variability, whereas classifying absences above a full-day threshold (≥ 8 hours) is more feasible.

III. Methodology

1. Regression Analysis (Continuous Prediction)

We model absenteeism time in hours as a supervised regression problem using the features stated above. Because the target variable is strongly right-skewed, we apply log transformation during training and convert the predictions back to hours for evaluation and interpretation.

To prevent leakage across repeated records from the same employee, we split the data by employee ID. We reserved 20% of employees as a final test set and used the remaining 80% for training. Within the training set, we use 5-fold GroupKFold cross validation, again grouped by employee ID.

We then compared two supervised models, **Ridge Regression** as a regularized linear baseline and **XGBoost Regression** as a nonlinear tree-based model. For Ridge, numeric features are standardized and categorical features are one-hot encoded. For XGBoost, numeric features are left unscaled and categorical features are one-hot encoded. Ridge hyperparameters are selected with RidgeCV, and XGBoost hyperparameters are tuned with RandomizedSearchCV using the same grouped cross-validation design.

The model performances are evaluated mainly with **MAE**, since it is easy to interpret in hours and is less sensitive to outliers. We also report **RMSE** and $\mathbf{R^2}$ on the held-out test set.

2. Classification Strategy (Risk Threshold Identification)

We also built a binary classification model **XGBClassifier** to predict whether an absence lasts at least 8 hours. Because the classes are imbalanced between short-term (n = 425) and long-term (n = 271) absences, we used scale_pos_weight. This ensures the model assigns greater importance to correctly identifying long-term cases.

The shift from regression to binary classification was driven by the inherent nature of the data. Short-term absences are dominated by routine medical visits, while long-term absences are more closely related to physical or structural factors. Because of this, classification may be more informative than exact-hour regression.

The classification model's performance is measured primarily by **AUC-ROC**, because it provides a robust measure of the model's ability to discriminate between short-term and long-term absence. We also evaluated the model using **accuracy**, **precision**, **recall**, and **F1-score**.

3. Cluster Analysis

To identify distinct absenteeism personas grouping, we applied **K-Means** clustering on employee-level aggregated data, resulting in 36 profiles defined by 16 features. These included behavioral metrics (e.g., total hours, frequency), health indicators (e.g., BMI, age), and workplace/lifestyle factors (e.g., commute distance, workload, dependents). Identifiers and categorical labels (such as medical reasons) were excluded to avoid bias, and all features were normalized using StandardScaler.

Because K-Means is an unsupervised algorithm and cannot explicitly rank the features it used to form its clusters, a **Random Forest Classifier** was trained as a supervised proxy to predict the newly generated K-Means cluster assignments, we reverse-engineered the feature importance scores to determine which variables truly drove the workforce segmentation.

IV. Experiment Setup and Analysis Results

1. Regression Analysis

After preprocessing, the supervised learning dataset was split by employee ID into training and test sets. The training set contained 494 records from 26 employees, and the test set contained 202 records from 7 employees. The final feature set included 20 predictors (excluding the employee ID).

For Ridge Regression, we used standardized numeric variables, one-hot encoded categorical variables, and selected the regularization strength with RidgeCV. For XGBoost, we used one-hot encoded categorical variables and tuned hyperparameters with RandomizedSearchCV over 50 sampled parameter settings using 5-fold GroupKFold. The model performance is shown below.

Overall, both models show that the available features contain some predictive signal, but absenteeism time remains difficult to predict accurately (see Table 2). XGBoost performs slightly better than Ridge, suggesting that limited nonlinear patterns may exist, though the improvement was modest. The relatively low R-squared values also indicate that much of the variation in absenteeism hours is likely driven by factors not captured in the dataset.

Table 2. Regression Model Performance

Model	Average Cross-Validation MAE	Test MAE	RMSE	R^2
Ridge Regression	4.90	4.49	11.23	0.074
XGBoost Regression	4.85	4.21	11.19	0.079

2. Classification Strategy - Risk Threshold Identification

To comprehensively evaluate the predictive power of our features, we conducted the classification analysis under two distinct scenarios:

Scenario 1 - Integrative Model (Including Absence Reasons): This model includes absence reasons and achieves the highest predictive performance (AUC = 0.905). Hyperparameters were first tuned using RandomizedSearchCV and further refined through manual adjustment. Results show reason-related features provide strong predictive power for identifying long absences.

- **Predictive Filtering of Short-Term Absences:** *Medical Visits* and *Other Medical* categories act as strong negative predictors, as they are mostly associated with short-duration absences (2-4 hours).
- **High-Impact Operational Risks:** *Injury* and *Musculoskeletal* conditions are the primary positive predictors of long absences. Although less frequent, they are more likely to result in full-day or multi-day absence.
- **Socio-Demographic & Lifestyle:** Socio-demographic factors such as *Son* (children) and *Pet* are associated with long absences, suggesting domestic responsibilities may influence likelihood of a full-day absence. Lifestyle indicators such as *Social Drinker* and *Social Smoker* also show predictive importance, although their effects are more indirect.

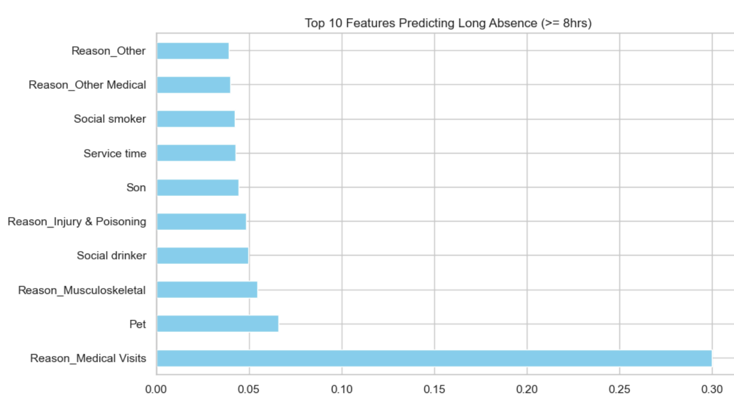

Figure 2. Top features for long-absence prediction with absence reasons

Scenario 2 - Behavioral Forecast (Excluding Absence Reasons): To evaluate the predictive power of socio-demographic factors alone, we built a model excluding absence reasons. It achieves an AUC of 0.812, indicating that demographic and work-related variables still provide meaningful predictive signals for long absences.

- **The Structural Anchors**: Variables such as number of children (*Son*), pet ownership (*Pet*), and *Service time* are associated with long absences. These factors may reflect greater domestic responsibilities, which increase the likelihood of taking full-day leave rather than short absences.
- **Lifestyle & Commute Inefficiency:** *Cost_per_km* shows strong predictive importance, suggesting that longer or more costly commutes are associated with a higher likelihood of long absences. Lifestyle indicators such as *Social Drinker* and *Social Smoker* also contribute to prediction, although their effects are less direct.
- **Complexity of Aging**: The importance of *Age_Group_Sq* suggests a non-linear relationship between age and absence duration, indicating that long-absence risk may vary across different life stages.

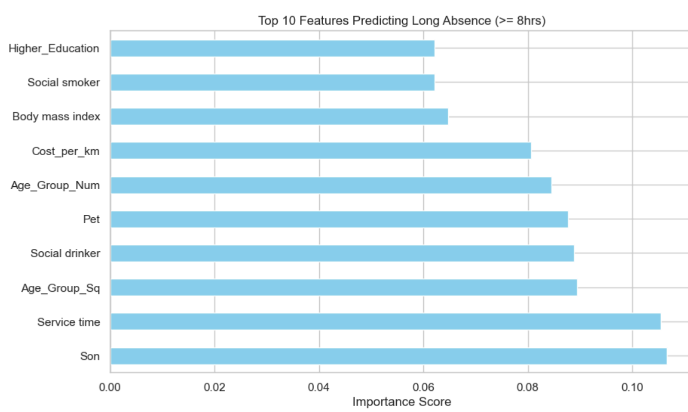

Figure 3. Top features for long-absence prediction without absence reasons

The model with reasons outperformed the model without reasons across all metrics. By comparing these two models, we can measure the extent to which behavioral patterns can act as proxies for medical drivers in predicting operational disruptions.

Table 3. Classification Model Performance

Scenario	AUC	Accuracy	Precision	Recall	F1-Score
With Reasons	0.905	0.84	0.77	0.81	0.79
Excluding Reasons	0.812	0.76	0.65	0.72	0.68

3. Cluster Analysis

To define the best K, we applied the Inertia (Elbow) Method and Silhouette Score. K=4 was selected to balance mathematical distinctness with actionable HR insights. The clustering algorithm successfully partitioned the workforce into four distinct, actionable profiles:

- **Cluster 0: The Educated Baseline (n=6):** Represents the lowest overall absenteeism (9.0 records, 55.6 total hours). This highly educated group experiences very short commutes (14.0 km) and the lowest lifestyle risk factors.
- **Cluster 1: The Young Commuters (n=6):** Characterized by extreme commute distances (41.3 km); more than double the workforce average. They are the youngest demographic (avg. age 29.5) and are prone to frequent, but highly brief, absences.
- **Cluster 2: The Severe Medical Cases (n=9):** This group exhibits the lowest absence frequency (7.6 records) but the highest average duration per event (12.0 hours). Comprising older employees (avg. age 41.5), this profile captures rare but severe extended medical leaves.
- **Cluster 3: The High-Risk Frequent Flyers (n=12):** The critical target for HR intervention. This group dominates the metrics with massive absence rates (40.0 records, 284.6 total hours), alongside the highest baseline BMI (28.8) and daily workloads.

Feature importance from the Random Forest surrogate model (see Figure 4) shows that total_records (0.106) and total_hours (0.102) are the primary drivers of clustering, followed by structural and health factors such as service_time and weight. In contrast, lifestyle variables (smoker, drinker) have negligible impact. This indicates that the four personas are primarily defined by absence severity and baseline health, rather than individual lifestyle choices.

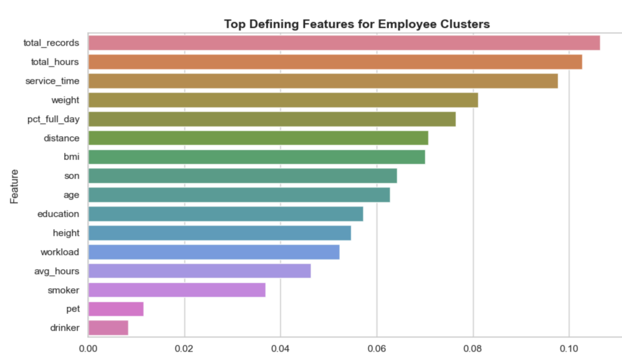

Figure 4. Top Defining Features for Employee Clusters

V. Conclusion and Future Work

The regression results showed that XGBoost Regressor performed slightly better than Ridge Regression, with a lower test MAE, although the improvement was modest. Both models had low R^2 values on the held-out test set, suggesting that while the available features contain some predictive information, exact absenteeism hours remain difficult to predict accurately. In contrast, the XGBClassifier showed that long-term absenteeism can be predicted with relatively high accuracy, especially when absence reasons are included. This indicates that although exact duration is difficult to model, the risk of full-day absence is driven by more stable and identifiable structural patterns.

The K-Means clustering analysis successfully shifts the analytical focus from isolated absence events to holistic employee personas. By segmenting the workforce into four distinct profiles, the model proves that severe absenteeism is not random, but deeply tied to structural factors like baseline health, commute friction, and tenure. This insight empowers HR to abandon reactive, one-size-fits-all policies in favor of highly targeted, proactive interventions, ultimately improving both operational workforce planning and employee well-being.

Overall, the results suggest that although exact absenteeism duration is difficult to predict, combining classification and clustering provides more meaningful insights into absenteeism patterns and long-absence risk.

VI. References

Llamas Blázquez, Pablo. “Predicting Workplace Absenteeism Using Machine Learning: A Pilot Study in Occupational Health.” *Journal of Occupational Medicine and Toxicology*, vol. 20, no. 1, 2025, p. 38. Crossref, doi:10.1186/s12995-025-00482-5.

Martiniano, Andréa, et al. “Application of a Neuro Fuzzy Network in Prediction of Absenteeism at Work.” *7th Iberian Conference on Information Systems and Technologies (CISTI 2012)*, 2012, pp. 1-4.

Skorikov, Mikhail, et al. “Prediction of Absenteeism at Work Using Data Mining Techniques.” *2020 5th International Conference on Information Technology Research (ICITR)*, 2020, pp. 1-6.

Wahid, Zaman, et al. “Predicting Absenteeism at Work Using Tree-Based Learners.” *Proceedings of the 3rd International Conference on Machine Learning and Soft Computing*, 2019.